AI-102 Natural Language Processing (NLP)

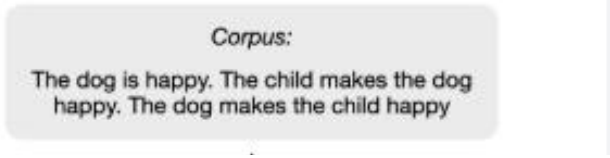
Dr. Jimmy Ardiansyah

EMERGING & COLLABORATIVE STUDIES

Feature Engineering for NLP

Bag of Words (BoW)

"Bag of Words" (BoW) is a fundamental method used in natural language processing (NLP) for text representation. It converts text into numerical features by treating each document as an unordered collection of words, ignoring grammar, word order, and context, but retaining the frequency of each word. it involves three main steps



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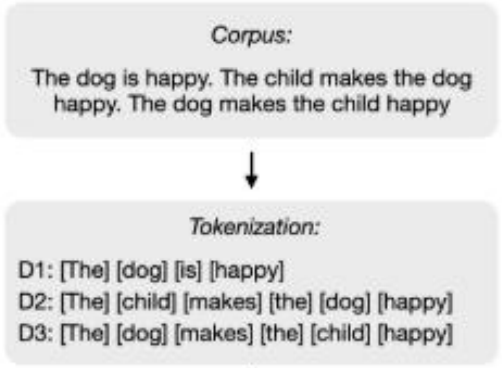
Tokenizing the Text : This is the initial step where text is broken down into individual words or "tokens." For example, the sentence "*'The dog is happy. The child makes the dog happy. The dog makes the child happy'*" would be tokenized into [“The", “Dog", "is", “happy"]. Tokenization is crucial for identifying the basic units that will form the vocabulary.



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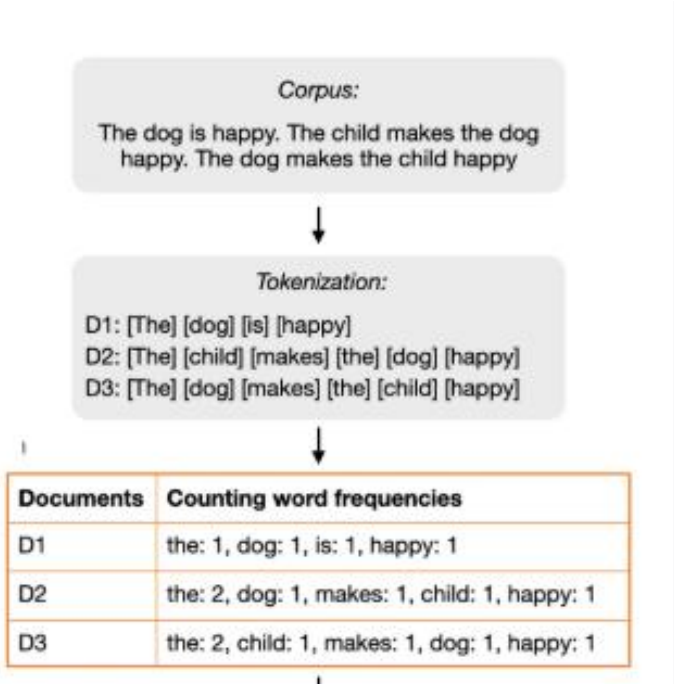

Documents	Counting word frequencies
D1	the: 1, dog: 1, is: 1, happy: 1
D2	the: 2, dog: 1, makes: 1, child: 1, happy: 1
D3	the: 2, child: 1, makes: 1, dog: 1, happy: 1

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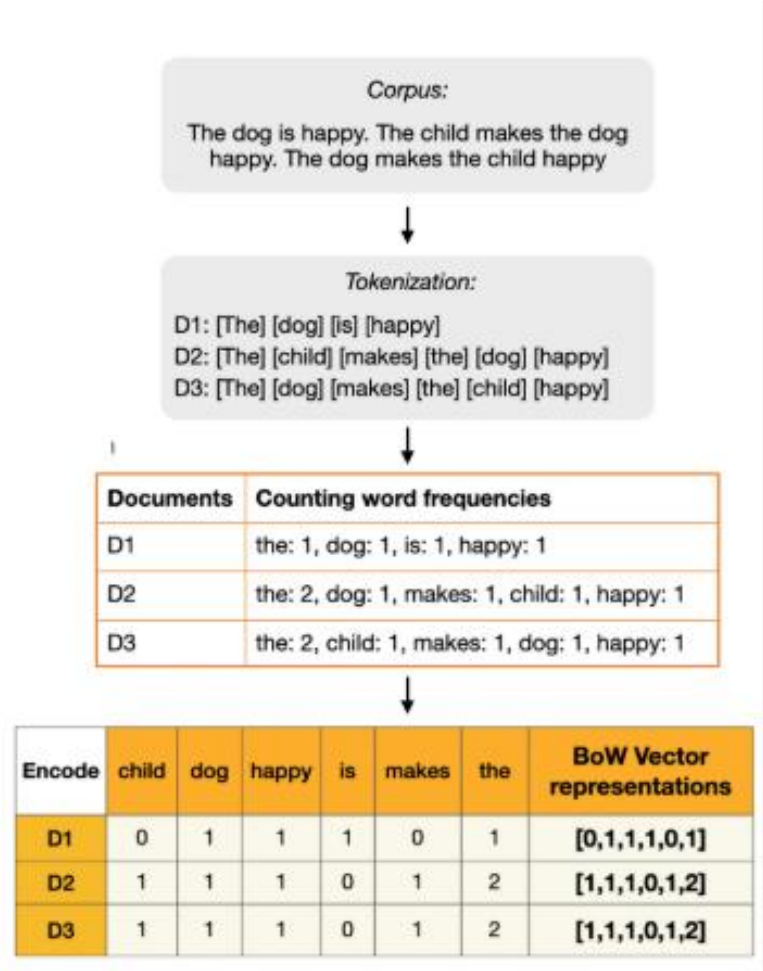
Building a Vocabular: After tokenization, a unique set of words from the entire text corpus is compiled to create the vocabulary.

Vectorizing the Text: In this final step, each document is converted into a numerical vector based on the established vocabulary.



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Encode	child	dog	happy	is	makes	the	BoW Vector representations
D1	0	1	1	1	0	1	[0,1,1,1,0,1]
D2	1	1	1	0	1	2	[1,1,1,0,1,2]
D3	1	1	1	0	1	2	[1,1,1,0,1,2]

Advantages and Limitation Bag of Words

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- Simplicity: BoW is "remarkably straightforward and easy to understand," making it highly accessible and allowing for "quick adoption and experimentation.“
- Efficiency: It offers "computational efficiency," especially for "small to medium-sized text corpora," leading to faster processing times.
- Baseline: BoW often serves as a "robust baseline for more complex models," providing a foundational benchmark for comparison with advanced techniques.

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- Loss of Context: A significant drawback is its "neglect of the order and context of words." This means "the sequence of words often plays a crucial role in conveying the true meaning of the text," which BoW fails to capture, potentially missing "important nuances.“
- High Dimensionality: As the vocabulary size increases with larger text corpora, BoW can lead to "high-dimensional feature vectors," making the model "cumbersome and difficult to manage.“
- Sparsity: The generated feature vectors are often "sparse," meaning "most elements in these vectors are zero." This "can be inefficient to process and may require additional computational resources and techniques to handle effectively."

Advantages and Limitation Bag of Words

```python
import pandas as pd
from sklearn.feature_extraction.text import CountVectorizer

# 1. Sample Text Data
# A simple list of documents (sentences) for our demonstration.
documents = [
    "The quick brown fox jumps over the lazy dog.",
    "A lazy cat naps on the warm sun.",
    "The sun is warm and bright."
]

# 2. Create the Bag of Words Model
# CountVectorizer handles tokenization, building the vocabulary, and counting word
# It automatically converts text to lowercase and removes punctuation.
# Stop words (common words like "the", "a", "is") are optional but often useful to
vectorizer = CountVectorizer(stop_words='english')

# 3. Fit and Transform the Data
# 'fit' learns the vocabulary from the documents.
# 'transform' converts the documents into a matrix of token counts.
bow_matrix = vectorizer.fit_transform(documents)

# 4. Inspect the Output
# Get the vocabulary (unique words) that the model learned.
vocabulary = vectorizer.get_feature_names_out()

# Convert the sparse matrix to a dense DataFrame for easier viewing and interpreta
# This makes the word counts for each document visible.
bow_df = pd.DataFrame(bow_matrix.toarray(), columns=vocabulary)

# Print the results
print("--- Our Vocabulary (Learned from the documents) ---")
print(vocabulary)
print("\n--- The Bag of Words Matrix (Document-Term Matrix) ---")
print(bow_df)
print("\n--- Shape of the matrix (documents x words) ---")
print(bow_df.shape)
```

```
--- Our Vocabulary (Learned from the documents) ---
['bright' 'brown' 'cat' 'dog' 'fox' 'jumps' 'lazy' 'naps' 'quick' 'sun'
 'warm']

--- The Bag of Words Matrix (Document-Term Matrix) ---
   bright  brown  cat  dog  fox  jumps  lazy  naps  quick  sun  warm
0       0      1    0    1    1      1     1     0      1    0     0
1       0      0    1    0    0      0     1     1      0    1     1
2       1      0    0    0    0      0     0     0      0    1     1

--- Shape of the matrix (documents x words) ---
(3, 11)
```

TF-IDF (Term Freq.-Inverse Document Freq.)

TF-IDF is a numerical statistic used in NLP to reflect the importance of a word to a document within a corpus. Unlike the Bag of Words model, which only counts word occurrences, TF-IDF provides a more sophisticated representation by considering the significance of each word in relation to the entire text collection.

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Term Frequency (TF): This is the simple part—just the count of how many times a word shows up in a single document. A word like "apple" appearing 5 times in a document has a higher TF than a word like "banana" that only appears once.

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Inverse Document Frequency (IDF): This is the clever part. It measures how unique or rare a word is across all the documents.

- Common words like "the," "is," or "and" appear everywhere. Their IDF score will be very low, so they don't get much importance.
- Rare words like "photosynthesis" or "quantum" are very specific. They will have a high IDF score, which makes them more important.

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TF-IDF Score is the result of multiplying these two scores together: TF x IDF.

$$TF - IDF(t,d) = TF(t,d) \times IDF(t)$$

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Importance Weighting: TF-IDF assigns a higher value to words that are significant to a specific document, while downplaying the importance of common, noisy words like "the" or "is." This highlights the unique vocabulary of each document.

Improved Feature Representation: Unlike basic frequency counts, TF-IDF provides a more nuanced and informative representation of text. It balances how often a word appears in a document (Term Frequency) with how rare it is across the entire collection of documents (Inverse Document Frequency).

Versatility: TF-IDF can be applied to a wide range of NLP tasks, including information retrieval, text classification, and clustering. The numerical data it generates can be easily used as input for various machine learning models.

Enhanced Performance: By providing a better representation of text, TF-IDF often improves the performance of machine learning models in classification and clustering tasks, especially when dealing with large datasets where distinguishing important terms is critical.

Ease of Use: TF-IDF is easy to implement using popular Python libraries like scikit-learn, making it accessible for both practitioners and researchers.

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Ease of Use: TF-IDF is easy to implement using popular Python libraries like scikit-learn, making it accessible for both practitioners and researchers.

Sparsity: Like the Bag of Words model, TF-IDF can produce high-dimensional and sparse feature vectors, especially with large vocabularies. This can lead to increased memory usage and computational inefficiency.

Context Ignorance: A significant drawback is that TF-IDF treats each word independently. It does not capture semantic relationships, word order, or the surrounding context, which is crucial for a deeper understanding of language. More advanced models like word embeddings are designed to overcome this limitation.

TF-IDF Example

We'll use three short documents:

Document 1: "The cat sat on the mat."

Document 2: "The dog chased the cat."

Document 3: "The dog and the cat."

Step 1: Calculate Term Frequency (TF)

Term Frequency is just the count of each word in each document. We'll normalize this by dividing the word's count by the total number of words in that document.

Word	Doc 1 (6 words)	Doc 2 (5 words)	Doc 3 (5 words)
the	$2/6 = 0.33$	$2/5 = 0.40$	$2/5 = 0.40$
cat	$1/6 = 0.17$	$1/5 = 0.20$	$1/5 = 0.20$
sat	$1/6 = 0.17$	0	0
on	$1/6 = 0.17$	0	0
mat	$1/6 = 0.17$	0	0
dog	0	$1/5 = 0.20$	$1/5 = 0.20$
chased	0	$1/5 = 0.20$	0
and	0	0	$1/5 = 0.20$

Step 2: Calculate Inverse Document Frequency (IDF)

Inverse Document Frequency measures how rare a word is across all documents.

Word	Documents containing the word	IDF Score
the	3	$\log\_e(3/3) = \log\_e(1) = 0$
cat	3	$\log\_e(3/3) = \log\_e(1) = 0$
sat	1	$\log\_e(3/1) = \log\_e(3) \approx 1.10$
on	1	$\log\_e(3/1) = \log\_e(3) \approx 1.10$
mat	1	$\log\_e(3/1) = \log\_e(3) \approx 1.10$
dog	2	$\log\_e(3/2) = \log\_e(1.5) \approx 0.41$
chased	1	$\log\_e(3/1) = \log\_e(3) \approx 1.10$
and	1	$\log\_e(3/1) = \log\_e(3) \approx 1.10$

Step 3: Calculate TF-IDF Score

Finally, we multiply the TF and IDF scores for each word in each document

Word	Doc 1	Doc 2	Doc 3
the	$0.33 \times 0 = 0$	$0.40 \times 0 = 0$	$0.40 \times 0 = 0$
cat	$0.17 \times 0 = 0$	$0.20 \times 0 = 0$	$0.20 \times 0 = 0$
sat	$0.17 \times 1.10 \approx 0.19$	$0 \times 1.10 = 0$	$0 \times 1.10 = 0$
on	$0.17 \times 1.10 \approx 0.19$	$0 \times 1.10 = 0$	$0 \times 1.10 = 0$
mat	$0.17 \times 1.10 \approx 0.19$	$0 \times 1.10 = 0$	$0 \times 1.10 = 0$
dog	0	$0.20 \times 0.41 \approx 0.08$	$0.20 \times 0.41 \approx 0.08$
chased	0	$0.20 \times 1.10 \approx 0.22$	0
and	0	0	$0.20 \times 1.10 \approx 0.22$

Summary:

The final TF-IDF scores show that words like **"the"** and **"cat"** have **no importance (a score of 0)** because they are so common. However, words that are unique to a document, like "**chased**" or **"sat"**, get a much higher score, indicating they are more relevant and important for understanding the specific content of that document.

TF-IDF vs. Bag of Words

Feature	Bag of Words (BoW)	Term Frequency-Inverse Document Frequency (TF-IDF)
Core Method	Counts the occurrences of each word in a document.	Calculates a weight for each word based on its frequency in a document and its rarity across the entire corpus.
Word Importance	Treats all words equally. High-frequency common words (e.g., "the") can dominate the representation.	Assigns higher weights to words that are important to a specific document, downplaying common words.
Representation	Simple and easy to implement.	More nuanced and informative, providing a context-aware representation.
Information Value	Can be skewed by words that carry little useful information about the document's specific content.	Highlights terms that are more relevant to the specific context of the document, leading to more precise analysis.
Suitability	Best for basic text analysis where word importance is not a critical factor.	Better suited for scenarios where understanding the significance of terms within the broader corpus is crucial.

Word Embedding Techniques

Word embeddings are a powerful NLP technique that maps words to vectors of real numbers. The primary goal is to create a representation where words with similar meanings have similar vector representations. This provides a dense and more informative alternative to traditional models like Bag of Words and TF-IDF, which often fail to capture nuanced semantic relationships.

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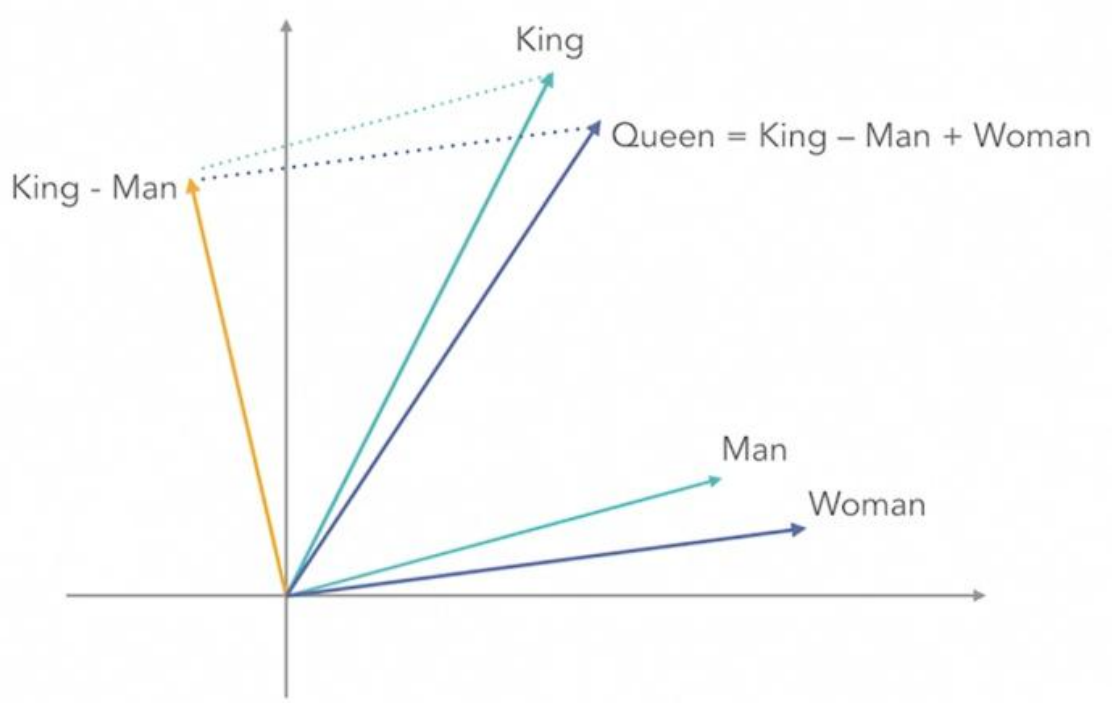
Continuous Vector Space: Words are represented as points in a continuous space, which allows for mathematical modeling of their relationships. A well-known example is the analogy *vector('king') - vector('man') + vector('woman')* resulting in a vector very close to *vector('queen')*.



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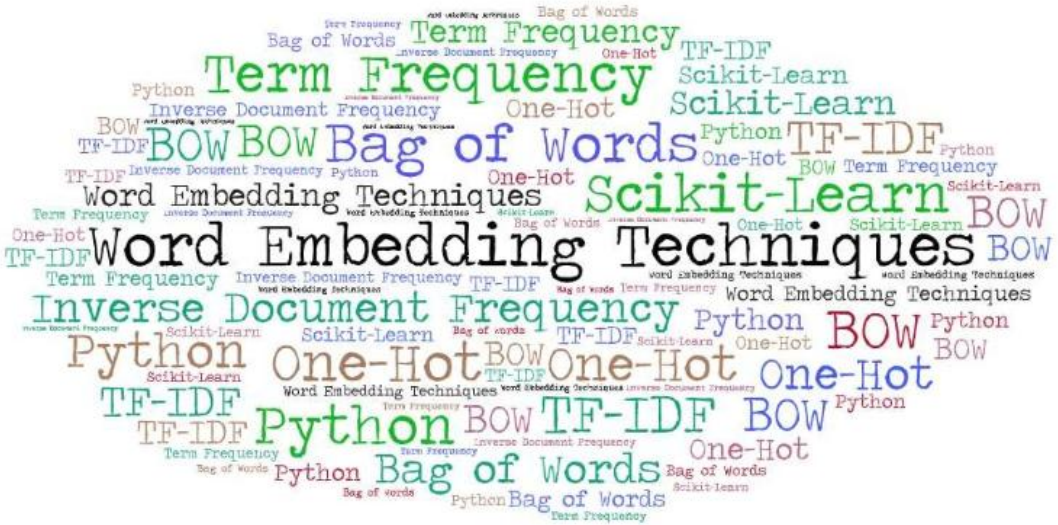
Transfer Learning: Pre-trained word embeddings, trained on massive text corpora, capture general linguistic properties. These can be applied to new NLP tasks (e.g., sentiment analysis, text classification) without needing to train a model from scratch, saving significant time and computational resources.



Word2Vec - Giving Words a Sense of Meaning

Word2Vec is a group of techniques in Natural Language Processing (NLP) used to create word embeddings. These are numerical representations of words (vectors) that capture their meaning and context. The core idea is that words that appear in similar contexts will have similar meanings and, therefore, will have similar vector representations.

Think of it like this: instead of a word being a standalone item, it becomes a point in a multi-dimensional space. Words with similar meanings will be located close to each other in that space. For example, the vector for "king" and the vector for "queen" would be very close, as would "man" and "woman." This allows computers to understand relationships between words.

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- **Continuous Bag-of-Words (CBOW):** This model predicts the current word based on its surrounding context words. For example, given the sentence **"The cat sat on the mat,"** the model might be trained to predict the word **"sat"** using the context words **"the," "cat," "on," and "mat."**

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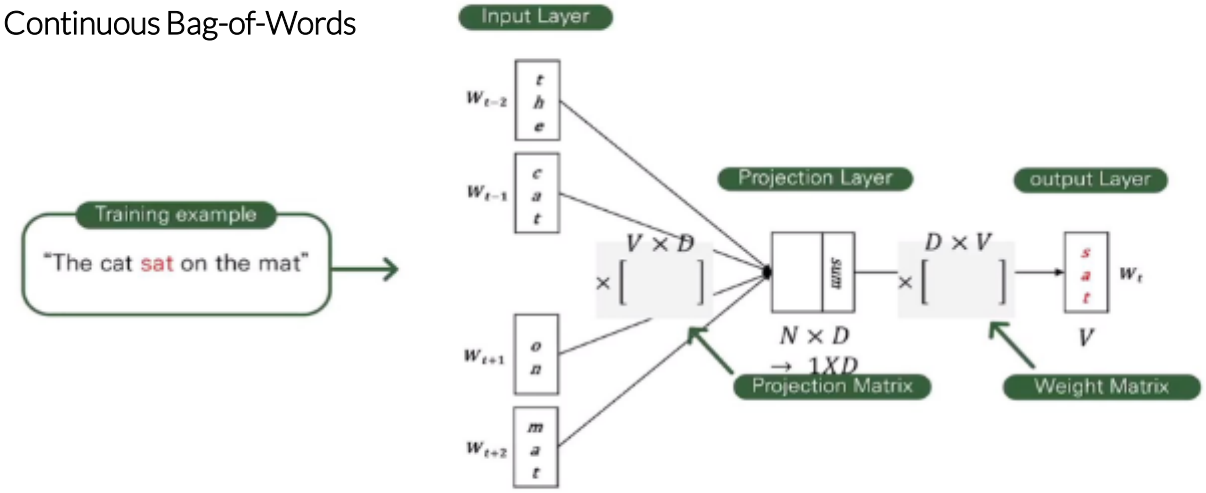

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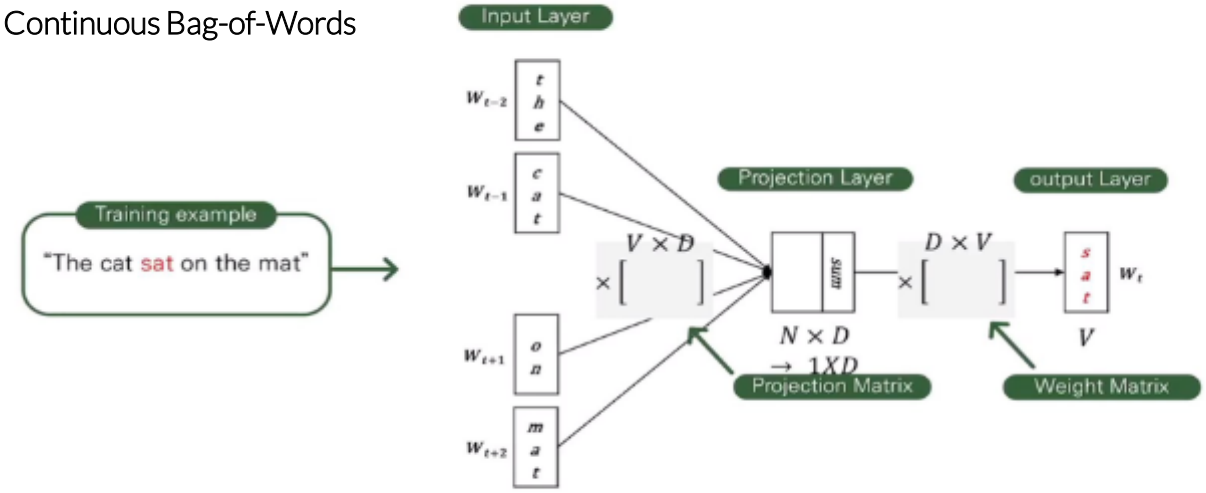
- **Skip-gram:** This model does the opposite. It takes a single word and tries to predict its surrounding context words. For example, given the word **"sat,"** the model might be trained to predict the words **"the," "cat," "on," and "mat."**

Continuous Bag-of-Words



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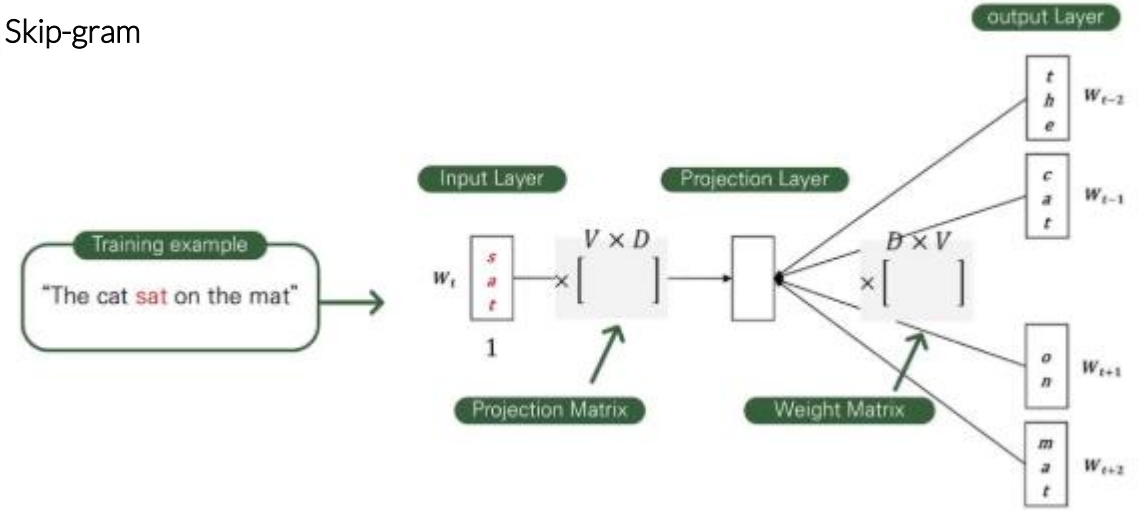

GloVe (Global Vectors for Word Representation)

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This method combines the best of two worlds:

- **Local Context Models :**These models learn from the local window of words.

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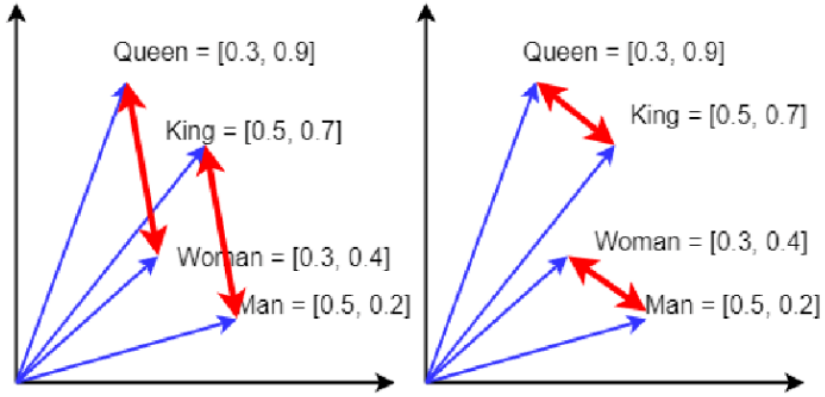

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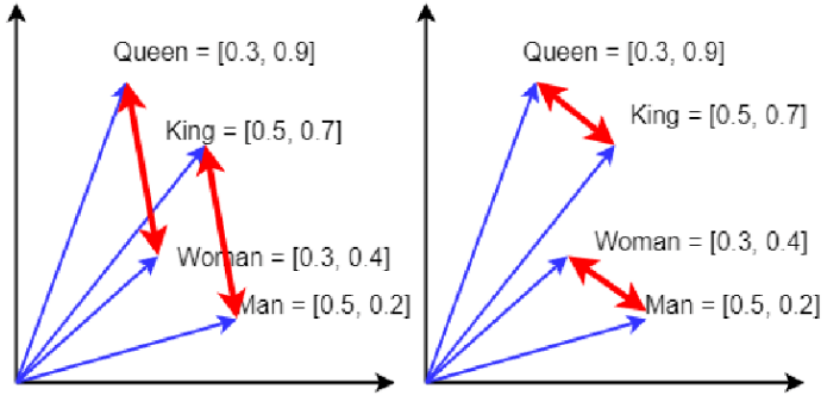

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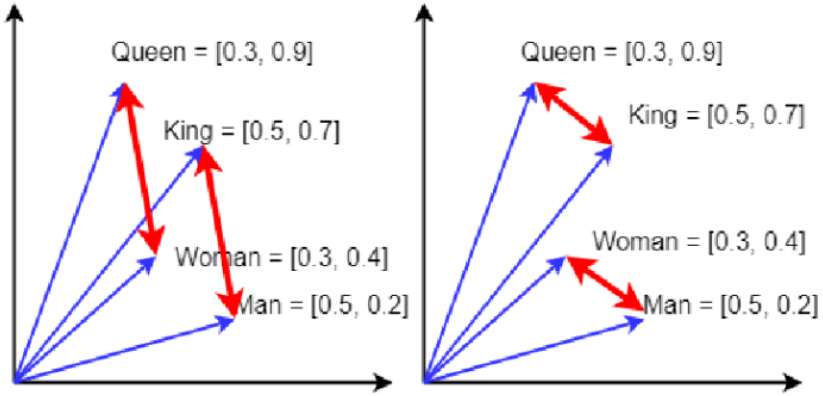

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Parallelism: The two red arrows are nearly parallel and have similar lengths. This visual similarity shows that the relationship between "King" and "Man" is very similar to the relationship between "Queen" and "Woman."

Therefore, by starting with the "King" vector, subtracting the "Man" vector (which removes the masculinity), and then adding the "Woman" vector (which adds femininity), you arrive at a new vector that is very close to the "Queen" vector. This works because the underlying word embeddings have learned to encode these abstract, relational meanings.

Word2Vec vs. GloVe

Feature	Word2Vec	GloVe (Global Vectors)
Core Method	Predicts words based on their immediate neighbors (local context). Uses neural networks (CBOW or Skip-Gram).	Performs matrix factorization on a global word-word co-occurrence matrix.
Context Focus	**Local:** Considers a limited window of words around a target word.	**Global & Local:** Captures statistical information from the entire corpus, considering both broad and immediate contexts.
Strengths	Highly efficient and fast to train on large datasets. Excels at capturing local contextual nuances.	Often produces more accurate embeddings for tasks benefiting from broader semantic relationships due to its global approach.

Introduction to BERT

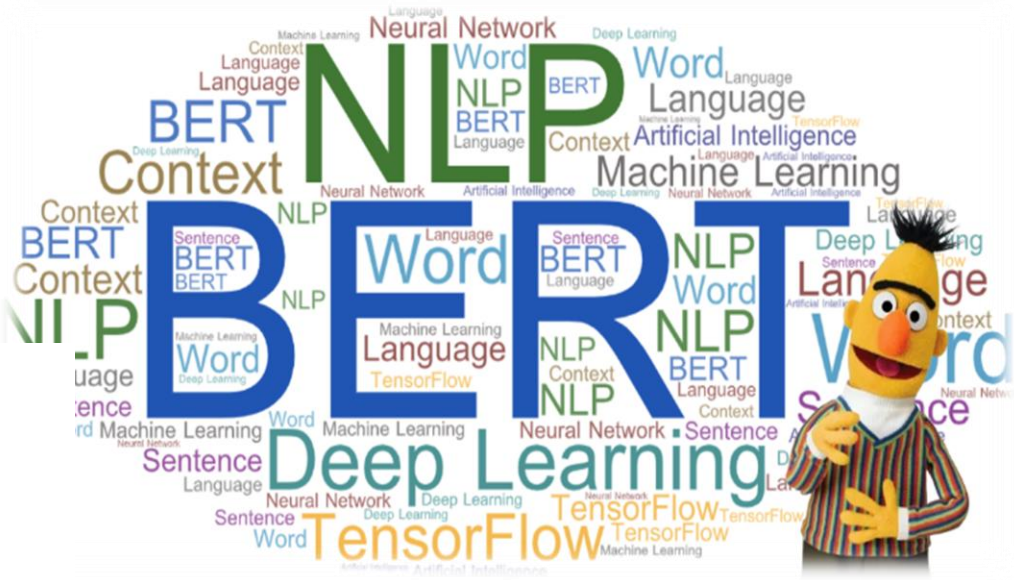
Earlier word embedding models like Word2Vec and GloVe created a single, static vector for each word. For example, the word "bank" would always have the same vector, regardless of its context. This is a problem because words often have multiple meanings.

Consider these two sentences:

"I need to go to the bank to deposit a check."

"The river bank overflowed after the heavy rain."

In the first sentence, "bank" refers to a financial institution, while in the second, it means the side of a river. Static embeddings fail to capture this **polysemy** (multiple meanings of a word), making them less effective for complex NLP tasks.



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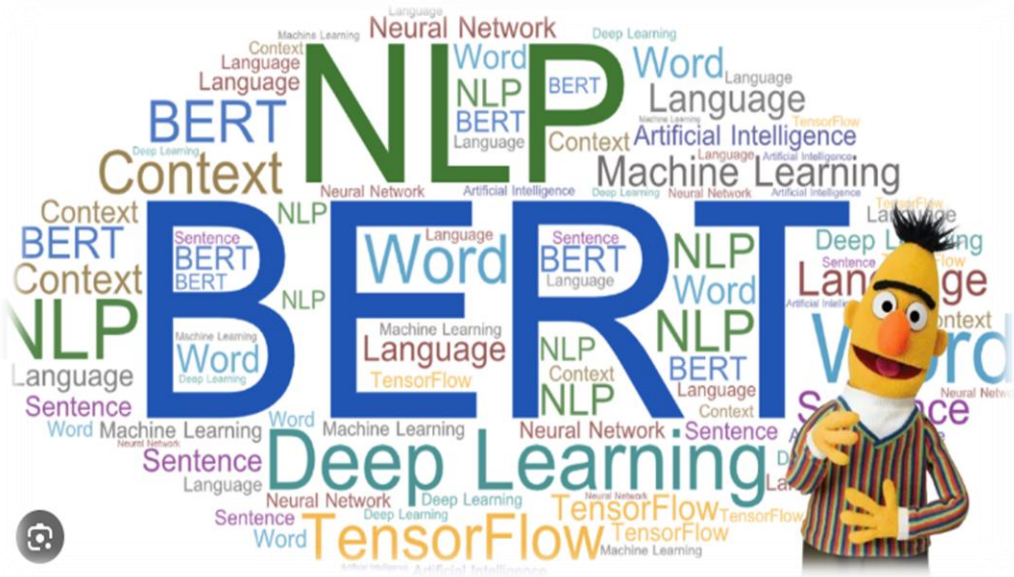

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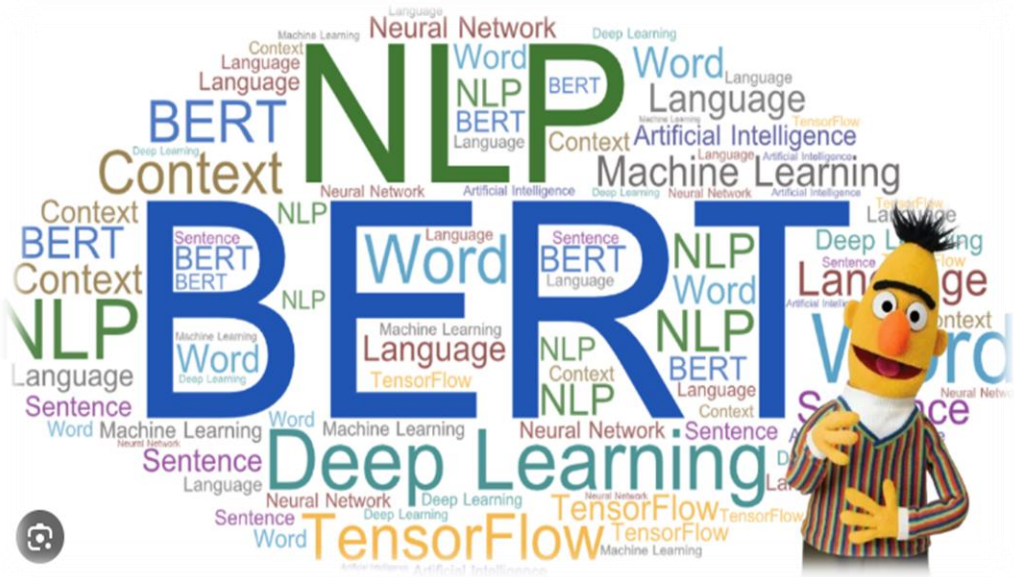
BERT's core innovation is its bidirectional nature. It reads a word in relation to all the other words in the sentence—both to its left and its right—at the same time. This is why it can understand the true meaning of a word in a specific context.



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- **Next Sentence Prediction (NSP):** The model is given two sentences, and it has to predict whether the second sentence logically follows the first one. This helps BERT understand how sentences relate to each other, which is crucial for tasks like question answering and conversation.

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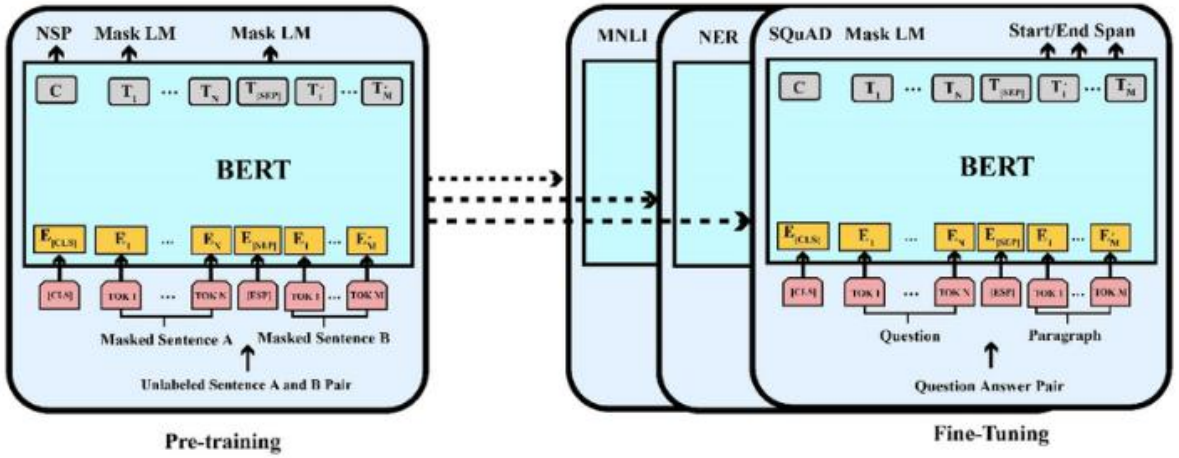
- **Process:** A new, small layer is added on top of the pre-trained BERT model. This new layer is designed specifically for the new task. For example, for a sentiment analysis task, this layer might be configured to output a "positive" or "negative" score.
- **Training:** The entire model (BERT plus the new layer) is then trained on a much smaller, labeled dataset. This is a quick process because BERT already has a powerful understanding of language from the pre-training phase. It only needs to make minor adjustments to its parameters to specialize in the new task. This approach is highly efficient because you don't have to train a new model from scratch for every new problem.



Advantages and Limitation BERT

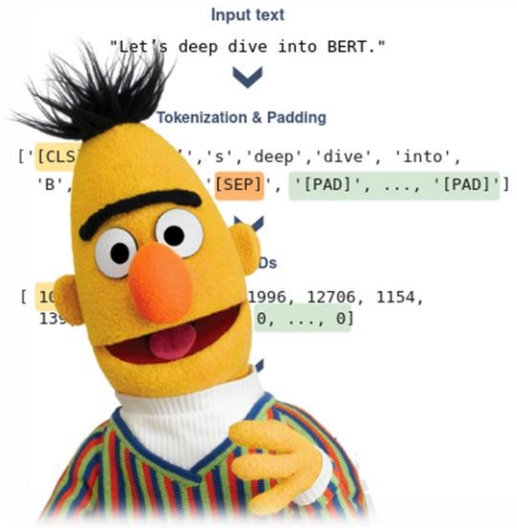
Context-Aware Embeddings: Provides a more nuanced and accurate representation of text by capturing the subtle meanings and relationships between words that traditional embeddings often miss.

State-of-the-Art Performance: Has set new industry standards by achieving top-tier performance on a wide range of NLP benchmarks and tasks, including question answering and sentiment analysis.

Transfer Learning: Its pre-trained models can be fine-tuned for specific tasks with relatively small amounts of labeled data, making it a highly versatile and efficient tool for various applications.

Computationally Intensive: BERT models are large and require significant computational resources for both training and inference, often necessitating access to high-performance hardware like GPUs or TPUs..

Complexity: The model's architecture and the fine-tuning process are complex, presenting a significant barrier for individuals new to NLP or those without a deep understanding of machine learning..

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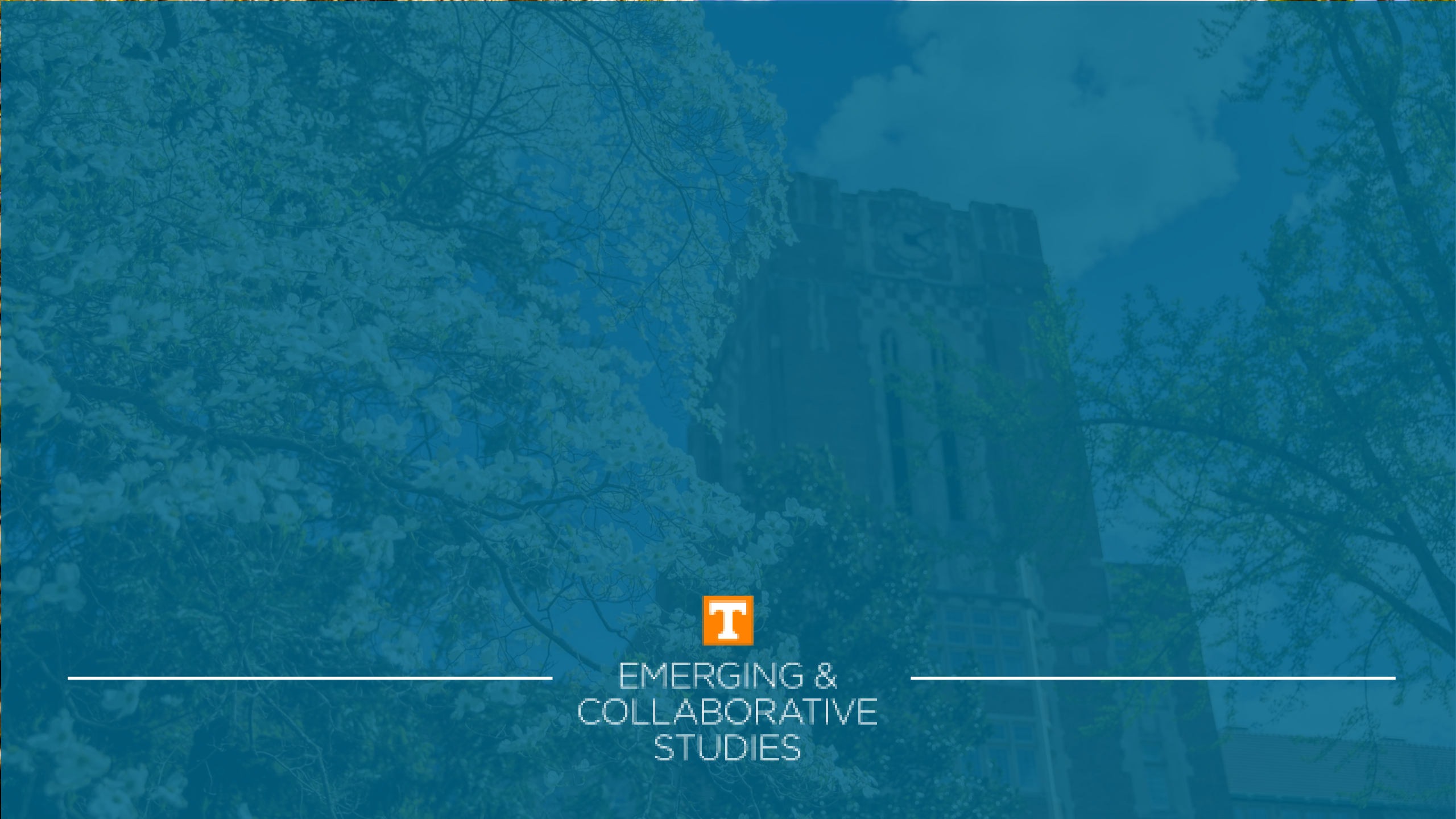
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